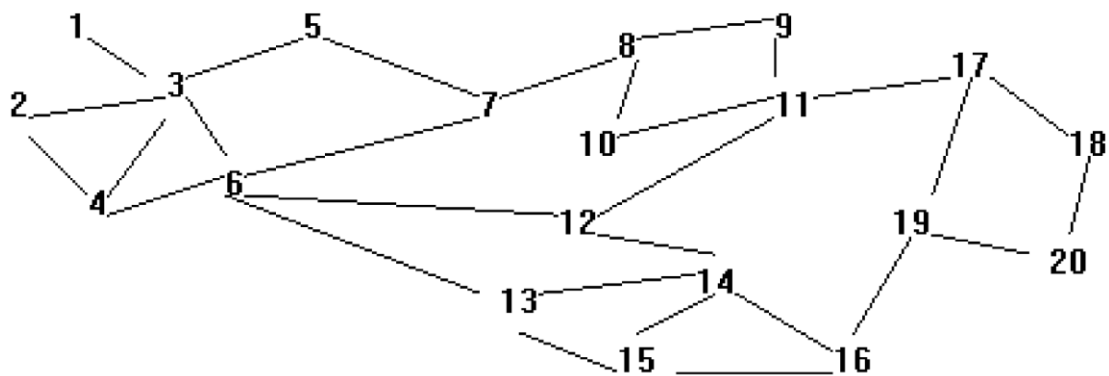


Q1. Risk

Risk is a board game in which several opposing players attempt to conquer the world. The gameboard consists of a world map broken up into hypothetical countries. During a player's turn, armies stationed in one country are only allowed to attack only countries with which they share a common border. Upon conquest of that country, the armies may move into the newly conquered country. During the course of play, a player often engages in a sequence of conquests with the goal of transferring a large mass of armies from some starting country to a destination country. Typically, one chooses the intervening countries so as to minimize the total number of countries that need to be conquered. Given a description of the gameboard with 20 countries each with between 1 and 19 connections to other countries, your task is to write a function that takes a starting country and a destination country and computes the minimum number of countries that must be conquered to reach the destination. You do not need to output the sequence of countries, just the number of countries to be conquered including the destination. For example, if starting and destination countries are neighbors, then your program should return one. The following connection diagram illustrates the first sample input.



Input:

Input to your program will consist of a series of country configuration test sets. Each test set will consist of a board description on lines 1 through 19. The representation avoids listing every national boundary twice by only listing the fact that country I borders country J when $I < J$. Thus, the I th line, where I is less than 20, contains an integer X indicating how many higher-numbered countries share borders with country I , then X distinct integers J greater than I and not exceeding 20 each describing a boundary between countries I and J . Line 20 of the test set contains a single integer ($1 \leq N \leq 100$) indicating the number of country pairs that follow. The next N lines each contain exactly two integers ($1 \leq A, B \leq 20; A \neq B$) indicating the starting and ending countries for a possible conquest. There can be multiple test sets in the input file; your program should continue reading and processing until reaching the end of file. There will be at least one path between any two given countries in every country configuration.

Output:

For each input set, the output is a single line contains 3 numbers, representing the starting country, destination country, and the smallest number of countries to be conquered. The numbers are

separated by blank spaces.

Sample input #1:

```
1 3
2 3 4
3 4 5 6
1 6
1 7
3 7 12 13
1 8
2 9 10
1 11
1 11
2 12 17
1 14
2 14 15
2 15 16
1 16
1 19
2 18 19
1 20
1 20
5
1 20
2 9
19 5
18 19
16 20
```

Sample output #1:

```
1 20 7
2 9 5
19 5 6
18 19 2
16 20 2
```

Sample input #2:

```
4 2 3 5 6
1 4
3 4 10 5
5 10 11 12 19 18
```

2 6 7
2 7 8
2 9 10
1 9
1 10
2 11 14
3 12 13 14
3 18 17 13
4 14 15 16 17
0
0
0
2 18 20
1 19
1 20
6
1 20
8 20
15 16
11 4
7 13
2 16

Sample output #2:

1 20 4
8 20 5
15 16 2
11 4 1
7 13 3
2 16 4